Day 136 | Unit 10 | Teacher plan

AASA Applications and Cumulative Review

Connect Rational Numbers, Expressions, and Ratios

Standards: 6.NS.A.1, 6.NS.C.7a, 6.NS.C.7b, 6.EE.A.2c, 6.RP.A.3a

Learning target: I can connect Rational Numbers, Expressions, and Ratios, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 14, 15, 24

Day 136 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Connect Rational Numbers, Expressions, and Ratios

Standards: 6.NS.A.1, 6.NS.C.7a, 6.NS.C.7b, 6.EE.A.2c, 6.RP.A.3a

Learning target: I can connect Rational Numbers, Expressions, and Ratios, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

Worked example 1

Calculate 1/2 ÷ 1/2. Simplify.

1. Rename the numbers as fractions: 1/2 = 1/2 and 1/2 = 1/2.

2. Keep the first fraction and multiply by the reciprocal of the second: 1/2 × 2.

3. Multiply numerators and denominators: (1 × 2) / (2 × 1) = 2/2.

4. Simplify by dividing top and bottom by 2: 1.

5. Check the result: 1/2 × 1 = 1/2, so the quotient is correct.

Answer: 1

Worked example 2

A recipe uses 5 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (26, 15)
B. (15, 9)
C. (10, 7)
D. (10, 6)

1. Multiply both values in a ratio by the same scale factor.

2. Equivalent ratios multiply both quantities by the same factor.

3. A uses 2

4. B uses 3.

5. C and D change only one scaled quantity.

Answer: B, D - (15, 9); (10, 6)

Day 136 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1/4 ÷ 1/2. Simplify.

02 / Constructed response

Compare -2 ___ -7 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Evaluate 4x + 3 for x = 4.

04 / Constructed response

A mix uses 3 cups of oats for 7 cups of seeds. Complete the table (oats, seeds): (3, 7), (6, ___), (12, ___).

05 / Constructed response

Calculate 1/2 ÷ 1. Simplify.

06 / Constructed response

Compare -3/2 ___ -7/5 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

A rectangle has length x + 8 and width 2. Use A = lw to write its area expression and find A when x = 5.

08 / Multiple select

A recipe uses 3 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (9, 12)
B. (6, 9)
C. (6, 8)
D. (16, 20)

09 / Multiple choice

Calculate 2/3 ÷ 1/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 10/3
B. 7/3
C. 3/10
D. 13/3

10 / Constructed response

Temperatures are -7/3°C and -5/2°C. Write an inequality comparing them and explain which is warmer.

Day 136 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Connect Rational Numbers, Expressions, and Ratios

Standards: 6.NS.A.1, 6.NS.C.7a, 6.NS.C.7b, 6.EE.A.2c, 6.RP.A.3a

Learning target: I can connect Rational Numbers, Expressions, and Ratios, and explain and check my reasoning.

1. 1/2

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

2. >

-2 > -7; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. 19

Substitute first: 4 × 4 + 3 = 19.

6.EE.A.2c | Constructed response

4. 14; 28

Multiply both parts by 2 and by 4.

6.RP.A.3a | Constructed response

5. 1/2

1/2 ÷ 1 = 1/2 × 1 = 1/2.

6.NS.A.1 | Constructed response

6. <

-3/2 < -7/5; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. A = 2(x + 8); 26 square units

Substitute x = 5: 2(5 + 8) = 26.

6.EE.A.2c | Constructed response

8. A, C - (9, 12); (6, 8)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

9. A - 10/3

2/3 ÷ 1/5 = 2/3 × 5 = 10/3.

6.NS.A.1 | Multiple choice

10. -7/3 > -5/2; -7/3°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response